A Bayesian Model for Lexical Availability of Chilean High School Students in Mathematics

Pedro Salcedo L.[1], Anita Ferreira C.[2], and Fernanda Barrientos C.[2]

[1] Research and Educational Informatics Department
[2] Department of Spanish Linguistics
University of Concepción, Chile
{psalcedo,aferreir}@udec.cl

Abstract. Lexical availability studies make an account from a statistical point of view about the lexicon of a group of people and the extent to which these might be considered representative of a bigger population. Recent findings in lexical availability prove the existence of patterns in order and word choice when different individuals recall words after a certain category or center of interest is given as a stimulus, and that these patterns can be modeled as a graph. The following article goes beyond the common idea of lexical availability in terms of frequency, stating that lexical availability can be modeled from a probabilistic perspective, i.e. calculating the probability that a certain word is available given other previous words. This work also has implications in prototype semantics theory, as it provides a mathematical model to obtain structures of the mental lexicon that can be extrapolated to the members of specific groups of people.

Keywords: Lexical availability, Bayesian networks.

1 Introduction

Although studies in lexical availability began in the 50's and were aimed to determine a basic vocabulary of French as a second language, nowadays the concept has been mainly researched and its scope and applications have varied significantly. An important application is in the field of education, from which many hypotheses have been posited in terms of its influence in learning different subjects and not only a second language. It has been stated that students with more available words in a certain field are more successful in their studies [7].

This work is part of a bigger research project (Fondecyt 1120911) [19] whose main goals are a) to quantify and describe the mental lexicon of high school students in the area of Mathematics regarding the following centers of interest: numbers (arithmetic), algebra, geometry, probabilities and chance (statistics) and other more common centers of interest (transportation, home and city); and b) to design a pedagogical proposal to generate an Adaptive Hypermedia System that allows for an increase in lexical availability in the centers of interest named above.

J.M. Ferrández Vicente et al. (Eds.): IWINAC 2013, Part I, LNCS 7930, pp. 245–253, 2013.
© Springer-Verlag Berlin Heidelberg 2013

2 Theoretical Framework

2.1 Studies in Lexical Availability

Mental Lexicon. The concept of mental lexicon refers to a mental state of the knowledge about words [6], a concept that attempts to explain the way that words are stored inside the human mind. Initially, the lexicon was understood as a static storage of different lexical items; however, as interest in the concept of mental lexicon increased in the fields of psycholinguistics and cognitive science, a more dynamic model emerged: today it is widely accepted that the mental lexicon is a complex group of networks consisting of lexical items that are interconnected and subject to change, as new words are learned, others are forgotten, and others are replaced [6]. In this respect, studies in lexical availability aim to measure the ease with which a word can be recalled by a specific person when asked about a certain category [8], which allows a better understanding of how words are connected to each other. However, more recent views of the mental lexicon do not only consider the connections that can be made among words from a certain category, but also the connections that can be made among words according to world knowledge [5].

Lexical Availability. Lexical availability studies began in the 50's, given the need to determine a basic vocabulary to teach French as a second language. Michéa (1953; cited in [4]) coined the term and defined it as a word that might not be particularly frequent, but is stored in the speaker's lexicon and can be recalled immediately when needed. Hence, the availability of a word can be measured as soon as a stimulus is presented to a certain individual. Usually, studies in lexical availability present a category (or centers of interest) to individuals who are members of a certain population and are then asked to mention or write down all the possible words that they can recall that are related to that category. The lists will show not only the words that individuals are able to associate to a certain category, but also the order in which they are recalled; hence, the availability of a certain word that showed up in first place in a list is not the same than another word in fifth place.

The resulting word lists are then analyzed with the formula proposed by López Morales (1978; cited in [4]) that was slightly adjusted to a more intuitive analysis by Echeverría et al [4]. The final formula is as follows:

$$D(p) = \frac{f_1 + \lambda f_2 + \lambda^2 f_3 + ... + \lambda^{n-1} f_n}{N}$$

However, it is worth mentioning that even though it is same person who performs this task, results can be different [6]. This is considered as a proof that the lexicon is mostly dynamic, and therefore the words are not stored in fixed places with a fixed recall time, or importance, or order of appearance.

Later, Echeverría et al. [3] considered that although these word lists obtained differed from each other, it was possible to find a pattern of lexical order. This

lead them to propose a lexicon model with the data obtained from their previous studies in lexical availability, in which the items of the different word lists were analyzed through a computational application. The program would form graphs in which the nodes represented the words and the edges had values that represented the number of times that a specific word was followed by another certain word. For instance, if the word uva (grape) was followed by the word durazno (peach) in two word lists, then the program would generate a graph with two nodes linked by an edge with a value of 2.

Our aim is to build on the model presented by Echeverría by adding a probabilistic component, thus being able to more accurately predict the mental structures in a particular group of people.

2.2 Bayesian Networks

When modeling environments or phenomena in which no direct causality relationships can be found, probability-based models can predict with a fair amount of precision the possible outcomes of an event. The Bayes Theorem calculates the probability that a certain event occurs when other previous events have occurred, by which it is possible to make an inference. A Bayesian network is a graphical model that finds probabilistic relationships among variables of a system [1] by using the Bayes formula.

According to graphs theory, a Bayesian network is a acyclic directed graph in which the nodes represent different variables and the edges represent the relationship among them, whether they are informational or causal [14]. The following is the simplest Bayesian network, in which A represents the parent node, and B is a child node. The edge pointing to B is interpreted as A being the previous event and B the second event, implying a causal relationship (A causes B).

Fig. 1. A simple Bayesian Network with one parent node (A) and one child node (B)

Therefore, a Bayesian network represents either the joint probability of a certain event to happen, given one or more previous events; or the probability of various events to happen, given one single previous event. In order to build a Bayesian network, some parameters are required: first, the prior probability of the event represented by the parent node; and secondly, the probability of A

given event B. After this, the probability of the presence of event A given B (i.e. $P(\frac{+a}{+b})$ can be calculated through the following formula:

$$P(X_1, .., X_n) = \prod_{i=1}^{n} P(\frac{X_i}{pa(X_i)})$$

Linguistics has been using Bayesian networks to model different aspects of human language, such as comprehension and production in general [12] or more specific aspects of it such as human sentence processing [11], and speech processing [10]. We attempt to understand semantic relationships among words as a probabilistic model, in which the order of available words is the subject of probability, as they appear only when others have been previously recalled.

2.3 Lexical Availability in Mathematics

Currently one of the most significant problems in mathematical education comes from the difficulty that students have in reading the wording in the discipline, which leads naturally to relate language to the process of learning of this science [17].

The act of communicating in "Mathematics" is related to a particular language, where every word must correspond to a communicative act to avoid errors in the construction of mathematical meanings ([13]; [20]; [15]; [2]; [16]).

It follows from the importance of knowing the mathematical lexicon latent in a particular group and model this lexicon so as to predict the probability that a word or construct appears according to the lexicon already acquired ([17]; [18]).

3 Bayesian Modeling of Lexical Availability

Echeverrías work went towards a model for the mental lexicon that could be representative of a certain group of people, supported by previous studies in lexical availability and graphs theory. The model that we propose takes Echeverras research as a framework, in order to model lexical structures that can be considered as representative of a certain group of people; in this case, the data corresponds to Chilean high school students and the center of interest is only related to Mathematics. However, our point of view includes a probabilistic component, which considers the existing lexicon in a certain group as well as its social background, which are described in the following section.

Our model calculates the probability that a certain word is available in a given populations lexicon given other variables: previous word(s) elicited, type of school where the student is enrolled, and sex of the student. As we already stated, the probability of a certain event when other previous events have also happened is given by the product of the product of such event given the probabilities of the parent nodes. Notice that in order to calculate the probability of a certain word that has been elicited after other word(s), then an iteration process is needed.

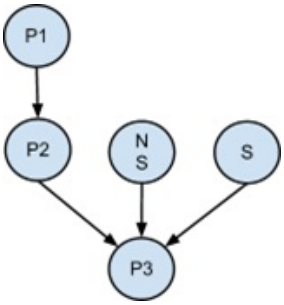

Fig. 2. A Bayesian model to predict the presence of a certain word in the lexicon of a specific propulation, given by P(p3 — pa(p3)) = P(p2 — p3) P(p3) P(p2) x P(NSa—p3) P(p3)P(NSa), etc

4 Methodology

To test the above model we used the data collected in the project FONDECYT 1120911 [17], for which we used the following methodology.

A test was applied in order to measure the latent lexicon, considering the student's social profile (sex, age and social background) which subsequently determine the frequencies of occurrence of each Word (node in the graph) and the frequency of relationships between each of them, allowing later analysis using the proposed model.

The test: This consisted of a form divided into two parts. The first was an identification item, in which subjects had to write down their name, age, sex and year of studies. In the second part of the test, subjects had two minutes for each center of interest to write down all the words that they could recall. .

Hypothesis: The proposed Bayesian model is effective at predicting the onset of lexical units in the students' semantic network, in the field of mathematics. .

Sample: It consisted of 1500 subjects, from three types of educational institutions in Concepcin, Chile (Municipal, fully funded by the government; Subvencionado, funded only in part by the government; and Particular, private), each of which has a specific social backgrounds. .

Variables: Since this study aims to test the proposed Bayesian model, it is necessary to consider the richness of the lexicon of every social background, thus considering as variables, sex, social class, and the number of words elicited by center of interest.

Sample edition:. In order to validate the data, the following decisions were taken: type all in caps; using singular form in nouns and adjectives; refer diminutives to the primitive word; and refer all verb forms to infinitive, except gerund and participle. Compound words were entered together by using hyphens.

To determine the words that belong to each center of interest, experts determined the words related to the progress maps in each area. Thus, all those words unrelated to mathematics and the specific center of interest were filtered.

Lastly, this study took only the first 50 words of the lexicon belonging to every social background and each specific center of interest with the largest number of connections (more than 5) and the highest frequency (greater than 8). This allowed nodes without significant influence on others to be left.

Data Processing: The data was computationally processed to calculate frequencies and probability. To determine the influences ELVIRA system was used (Fig. 3)

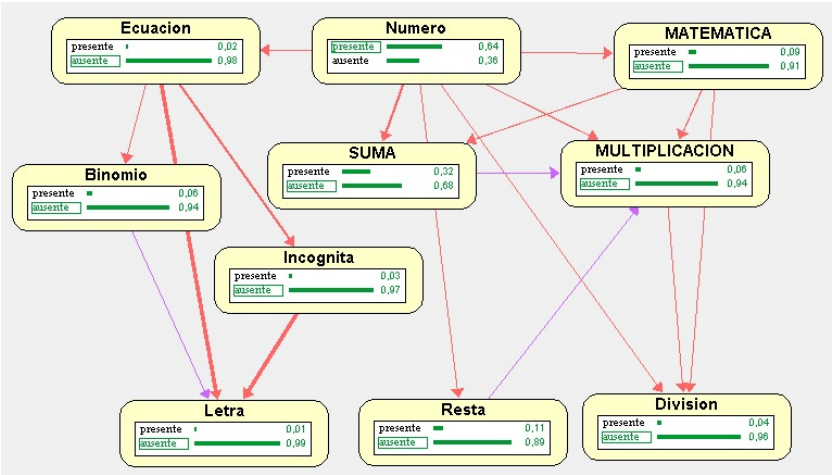

Fig. 3. Network developed with ELVIRA

5 Results

In order to test the model, only 10% of the individuals in the sample (150 students) was chosen, ignoring irrelevant nodes and edges (with a frequency lower than 8). The percentage that the probability of appearance of each node (word) shown by the model matches the words presented by the subject has been determined.

Table 1 shows an example of the first 10 words that were most mentioned by the students and the corresponding percentage, in the specific center of Algebra and considering social backgrounds from working class to upper class (Municipal, Subvencionado and Particular).

The frequency of the relationships among words is determined by the sequence in which students mentioned each word in the survey. Table 2 shows the number of times a word is mentioned after another.

Finally, to test the model percentage of certainty, a 10% of the whole sample has been used, which is compared with the prediction results of each of the words, obtaining table 3, which shows that on average between the three types of educational institutions, the model has a 65.

Table 1. The 10 most frequently mentioned words in the survey, with the corresponding relative frequencies

	MUNICIPAL	Fr Abs	Fr %	SUBENCIONADO	Fr Abs	Fr %	PARTICULAR	Fr Abs	Fr %
1	NUMERO	55	11,2%	NUMERO	159	6,6%	NUMERO	47	7,2%
2	LETRA	30	6,1%	LETRA	129	5,4%	LETRA	42	6,4%
3	MATEMATICA	24	4,9%	SUMA	114	4,7%	ECUACION	24	3,7%
4	SUMA	22	4,5%	RESTA	105	4,4%	MULTIPLICACION	23	3,5%
5	MULTIPLICACION	20	4,1%	MULTIPLICACION	99	4,1%	DIVISION	22	3,4%
6	DIVISION	18	3,7%	DIVISION	81	3,4%	RESTA	20	3,1%
7	RESTA	18	3,7%	MATEMATICA	73	3,0%	SUMA	19	2,9%
8	SIGNO	17	3,5%	ECUACION	68	2,8%	X	18	2,8%
9	PROBLEMA	13	2,7%	FRACCION	61	2,5%	INCOGNITA	17	2,6%
10	FRACCION	12	2,4%	BINOMIO	55	2,3%	FRACCION	13	2,0%

Table 2. Percentage of appearance of a word in a sequence

	DIVISION	FRACCION	LETRA	MATEMATICA	MULTIPLICACION	NUMERO	PROBLEMA	RESTA	SIGNO	SUMA
DIVISION	0,0%	0,0%	1,4%	0,0%	0,0%	0,0%	0,0%	0,7%	0,7%	0,7%
FRACCION	1,4%	0,0%	0,7%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%
LETRA	0,0%	0,0%	0,0%	0,7%	0,0%	0,0%	0,0%	1,4%	0,7%	6,1%
MATEMATICA	1,4%	0,0%	0,0%	0,0%	2,0%	2,7%	0,0%	0,0%	0,0%	0,0%
MULTIPLICACIC	0,7%	0,0%	1,4%	0,0%	0,0%	0,0%	0,0%	2,0%	0,0%	0,0%
NUMERO	0,0%	0,0%	0,7%	5,4%	0,0%	0,0%	0,0%	2,7%	0,0%	0,7%
PROBLEMA	0,0%	0,0%	0,0%	0,0%	1,4%	0,7%	0,0%	0,0%	0,0%	0,0%
RESTA	0,0%	0,0%	1,4%	2,0%	0,0%	1,4%	0,0%	0,0%	0,0%	0,0%
SIGNO	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	0,0%	2,0%
SUMA	2,0%	0,0%	2,7%	0,0%	0,7%	1,4%	0,0%	1,4%	0,0%	0,0%

Table 3. Percentage of appearance of a word in a sequence of appearance in the survey

	MUNICIPAL	SUBENCIONADO	PARTICULAR
DIVISION	68,0%	56,0%	74,0%
FRACCION	45,0%	58,0%	87,0%
LETRA	56,0%	64,0%	45,0%
MATEMATICA	68,0%	73,0%	67,0%
MULTIPLICACION	89,0%	78,0%	47,0%
NUMERO	67,0%	85,0%	84,0%
PROBLEMA	74,0%	69,0%	34,0%
RESTA	67,0%	56,0%	56,0%
SIGNO	54,0%	67,0%	67,0%
SUMA	64,0%	56,0%	75,0%
Promedio del %	65,2%	66,2%	63,6%

6 Conclusions

This work presented a Bayesian approach to model lexical availability in a certain population regarding a specific center of interest. Our work follows Echeverra's model of lexical availability [3] in which graphs display the relationships among the different words in a certain group's lexicon. The model was tested in a group of Chilean high school students in the field of Algebra, showing a 65% of accuracy in predicting the presence of a certain word.

This proposal has also implications in cognitive semantics, specifically in prototype theory. In particular, this model proves that certain lexical units are more likely to occur in a specific population regarding a center of interest.

Acknowledgements. Research Project - Fondecyt 1120911 Disponibilidad Léxica Matemática en estudiantes de Enseñanza Media y su aplicación en Hipermedios Adaptativos, National Commission for Scientific and Technological Research (CONICYT) of Chile.

References

1. Bashir, A., Khan, L., Awad, M.: Bayesian Networks. In: Wang, J. (ed.) Encyclopedia of Data Warehousing and Mining. Idea Group Publishing, Hershey (2005)
2. Dickson, L., Brown, M., Gibson, O.: El aprendizaje de las matemÃticas. Labor, Madrid (1991)
3. Echeverría, M., Vargas, R., Urzúa, P., Ferreira, R.: Dispografo: una nueva herramienta computacional para el análisis de relaciones semánticas en el léxico disponible. RLA Revista de Lingistica Teórica y Aplicada 46(1), 81–91 (2008)
4. Echeverría, M., Herrera, M., Moreno, P., Pradenas, F.: Disponibilidad léxica en educación media. RLA Revista de Lingistica Teórica y Aplicada 25, 55–115 (1987)
5. Elfman, J.: An alternative view of the mental lexicon. Trends in Cognitive Sciences 8(7), 301–306 (2004)
6. Ferreira, R., Echeverría, M.: Redes semánticas en el léxico dIsponible de inglés L1 e inglés LE. Onomázein 21(1), 133–153 (2010)
7. Giammatteo, M., Albano, H., Basualdo, M.: Competencia léxica y aprendizaje. Interrelación de resultados de investigaciones realizadas en los niveles medio y superior. Filología XXXIII 1-2, 31–57 (2000)
8. Hernández-Muñoz, N., Izura, C., Ellis, A.: Cognitive aspects of lexical availability. European Journal of Cognitive Psychology 18(5), 730–755 (2006)
9. Jurafsky, D.: A probabilistic model of lexical and syntactic access and disambiguation. Cognitive Science 20, 137–194 (1996)
10. McQueen, J., Huettig, F.: Changing only the probability that spoken words will be distorted changes how they are recognized. Journal of the Acoustic Society of America 131(1), 509–517 (2012)
11. Narayanan, S., Jurafsky, D.: Bayesian models of human sentence processing. In: COGSCI-1998, pp. 752–757. Lawrence Erlbaum, Madison (1998)
12. Narayanan, S., Jurafsky, D.: A Bayesian Model Predicts Human Parse Preference and Reading Time in Sentence Processing. In: Dietterich, T.G., Becker, S., Ghahramani, Z. (eds.) Advances in Neural Information Processing Systems, vol. 14, pp. 59–65. MIT Press, Cambridge (2002)

13. Orton, A.: DidÃctica de las matemÃticas. MEC y Morata, Madrid (1990)
14. Pearl, J.: Bayesian networks. Department of Statistics, UC Los Angeles (2011), http://escholarship.org/uc/item/5fx179hb (retrieved)
15. Pimm, D.: Speaking Mathematically. Routledge and Kegan Paul, New York (1987)
16. Rotherry, A.: Children reading mathematics. College of Higher Education, Worcester (1980)
17. Salcedo, P., Del Valle, M.: Disponibilidad Léxica Matemática en Estudiantes de Ensenanza Media de Concepción, Chile. In: XIV Evento Internacional Matecompu 2012. La Ensenanza de la Matemática, la Estadística y la Computación, Cuba, Universidad Pedagógica de Cuba, Matanzas (2012)
18. Salcedo, P., Nail, O.: Análisis de relaciones semánticas del léxico disponible en matemáticas en un hipermedio adaptativo. In: TISE 2012, Congreso Internacional de Informática Educativa, Santiago de Chile (Diciembre 2012)
19. Salcedo, P., Ferreira, A., del Valle, M., Nail, O.: Disponibilidad Léxica Matemática en estudiantes de Enseñanza Media y su aplicación en Hipermedios Adaptativos. In: Proyecto Fondecyt 1120911, de la Comisión Nacional de Investigación Científica y Tecnológica de Chile, en ejecución entre los años (2012)
20. Truran, K.: Children's understanding of random generators. Short oral communications. In: Proceeding of the XVIII International Conference for the Psychology of Mathematics Education. University of Lisbon (1994)